

Graph-aware VCF

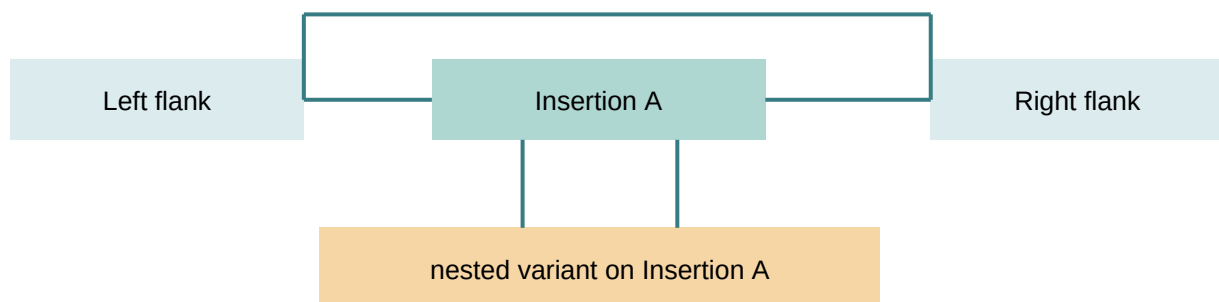
Encoding graph topology as variants

A practical guide to LinGraph grVCF: how to read its additional fields, interpret variants on insertions, and use the calls for graph reconstruction or standard VCF analysis.

The central idea

An insertion is both a variant and a named sequence. Its VCF **ID** can become the **CHROM** of another record, so SNPs, deletions, replacements, and further insertions can be described within the inserted sequence. An allele can also reuse oriented pieces of another allele rather than repeat its bases.

Parent coordinates encode where a variant attaches. Named sequence pieces encode what it shares. Together, these relationships describe graph topology while keeping event and sample information in a VCF-style table.



Insertion A adds an alternative route between reference flanks. The nested variant changes sequence within A. Worked examples are on page 5.

Choose the output for your task

Understand or exchange calls: keep grVCF, its headers, and required sequence catalogs.

Build graph topology: export rGFA with SNP, substitution, insertion, and deletion branches.

Use reference-based variant tools: convert representative alleles to standard VCF; keep graph-only records in the unplaced sidecar.

Trace an assembly through the graph: generate per-sample GAF walks using the graph, merged variants, and original sample mappings.

Updated 4 October 2026 | Replaces the September guide

These are LinGraph conventions, not a separate standardized VCF version. The current rGFA exporter includes all variant sizes by default.

01 / Read a record

grVCF uses a VCF 4.2-style envelope. INFO describes a representative event; FORMAT describes observations in sample/haplotype columns. Graph coordinates and custom values require a LinGraph-aware reader. [1, 2]

Column / concept	How to read it
CHROM	A reference/alternative locus or a parent variant ID. If it is an insertion ID, POS refers to that inserted sequence, not to the original chromosome.
POS / END	Event coordinates. SNP base positions and structural-variant breakpoints follow different conventions; see page 4.
ID	A stable variant identifier that may also name a sequence target and parent coordinate space. Keep dependencies when subsetting or renaming.
REF / ALT	SNPs have base alleles; structural events may use <INS>, , or <SUB>. Representative query sequence can be stored in INFO/SEQ.
QUAL / FILTER	Preserve emitted values. Export filters and required dependencies determine which branches are retained.
INFO / FORMAT	One representative versus per-observation evidence. A carrier can differ from the representative in size, sequence, or breakpoint.
Sample columns	Usually assembly haplotypes such as HG002_h1 and HG002_h2. These are not sequencing-depth measurements.

Genotypes and multiple observations

GT=1 means the haplotype supports the variant; **GT=0** is callable reference; **GT=.** is no call. Never turn missing data into reference support. Merging mode and available mapping coverage determine reference calls.

Read each FORMAT name from column 9, then split sample cells on colons. Number=. observation fields contain comma-separated values; preserve correspondence across those lists. GT remains one state. Decode escaped characters only after splitting the structural delimiters.

A shifted carrier

```
Row POS: 100
FORMAT: GT:TYPE:SIZE:EXTENDGRAPHCIGAR:TEMPLATEOFFSET
Sample: 1:INS:8:>8=:5
```

This abbreviated example supports the row representative, with a member breakpoint displaced 5 bases to the right. TEMPLATEOFFSET shifts the observation breakpoint; it does not shift coordinates inside the inserted sequence. Full records also retain query coordinates and other fields. [2, 3]

02 / INFO fields

Number=1 unless noted. Field availability depends on the producing stage; the file header is the schema for that file. [2, 3]

Field	Type	Meaning
SVTYPE	String	Event type: INS, DEL, SUB, or SNP. Merging may decompose replacement events into components.
END	Integer	Event end/second anchor. A merged deletion can carry a cluster envelope; use representative SVLEN to recover its actual deletion span.
NSUP	Integer	Number of supporting sample/haplotype columns, not read depth.
SVLEN	Integer	Representative signed length: insertion positive, deletion negative.
MAXSIZE	Integer	Maximum insertion/deletion component size among cluster members, not necessarily the net length change.
SEQ	String	Representative query allele. Default final caller output is plain sequence; merged insertions may use named graph-encoded pieces. Dot means absent.
EXTENDGRAPHCIGAR	String	Representative extended graph alignment. Default final caller output removes inline sequence payloads.
PACCLASS	String	Representative pseudo-linear assignment class: primary or fulllocusdup.
PAMAP	Integer; Number=.	IDs of originating <code>##pseudoLinearMapping</code> entries. Interpret their Path values to distinguish location from sequence-source information.
ALTERNATIVECIGAR	String	Query-to-source-locus alignment for a full-locus duplication insertion; adjacent elected pieces are joined with &.

Location and sequence source are different

A `##pseudoLinearMapping` entry contains ID, Sample, Query, Reference, Category, and Path. Its coordinates are declared **0-based half-open**. **Path=pri** locates the haplotype bases; **Path=alt** describes a sequence-source layer and adds no extra bases to the haplotype. INFO/PAMAP cites these entries in caller output. Do not count an alt-source mapping as another copy of the haplotype. [2]

Mapping IDs are scoped to their source header. Merged output need not retain the caller mapping layer: current merging filters alt-only records and removes mapping-specific metadata. Keep original sample VCFs for assembly-walk reconstruction. [3, 6]

Older or intermediate fields

CIGAR, SVINSSEQ, SVDELSEQ, and QUERYSEQ occur in internal/older files. The final caller replaces their sequence representation with SEQ and EXTENDGRAPHCIGAR. Legacy HSV/HSNP packed sample payloads remain accepted by downstream readers.

03 / FORMAT and coordinates

FORMAT field	Number / Type	Meaning
GT	1 / String	Haploid variant/reference/no-call state.
TYPE / SIZE	. / String; Integer	Observation event type and signed size.
EXTENDGRAPHCIGAR	. / String	SV observation aligned against a representative or named target. Bare X operations may lack alternate bases.
BASE	. / String	Alternate base for SNP observations.
ASSEMBLYCONTIG	. / String	Source assembly contig; must match the supplied FASTA.
QUERYCOORD	. / String	0-based query coordinate or range plus strand. A minus-strand single-SNP coordinate is a traversal boundary; its base lies at coordinate minus one.
TEMPLATEOFFSET	. / Integer	Signed member-breakpoint displacement from row POS; positive right, negative left.
LIFTCATEGORY	. / String	GenomeLift category: Pri (types 1/2), Dup (type 3), DivergeDup (type 4), or .
ALLELENAME / LABEL_H	. / String	Optional PA allele/coordinate provenance. Contributors may be joined with &. Final caller drops these by default; --PAtag retains them.

Normalize coordinates before interpreting topology

Event / parent	Zero-based interval used by current rGFA exporter
SNP on any parent	[POS - 1, POS): SNP POS is a 1-based base coordinate.
INS/SUB on root	[POS, END); pure INS with END=POS has an empty replacement interval.
INS/SUB on insertion	Start = POS - 1 by default. Pure INS ends at start; nonempty replacement ends at END.
DEL	Start follows the root/nested breakpoint convention; end = start + abs(SVLEN). Only legacy rows without SVLEN use END.

These are the cohort writer/exporter conventions, not a universal interpretation of symbolic VCF. --nested-pos-base 0 is only for inputs using zero-based nested breakpoints. A D inside an insertion alignment consumes template sequence; it is not automatically a deletion on the chromosome. [4]

04 / Variants on insertions

Teaching examples below show selected columns only. They omit headers and sample observations and are not a complete input file. Each child illustrates an independent allele on the same eight-base insertion.

```
CHROM POS ID REF ALT INFO
chr1 100 A N <INS> SVTYPE=INS;END=100;SVLEN=8;
      SEQ=ACGTTGCA
A      5 B N <INS> SVTYPE=INS;END=5;SVLEN=2;SEQ=GG
A      3 S G A SVTYPE=SNP
A      3 D N <DEL> SVTYPE=DEL;END=4;SVLEN=-2
```

The parent ID creates a local coordinate space

A inserts ACGTTGCA after root base 100. **B** uses CHROM=A and inserts GG at boundary 4 (POS-1), producing **ACGTGGTGCA** within the outer chromosome context.

S replaces the third base of A, G, with A, producing **ACATTGCA**.

D starts at boundary 2 and deletes two bases, producing **ACTGCA**. The reference route through A remains available.

A further variant can use B as its parent, so the same representation supports deeper nesting. These examples do not imply that every child occurs in the same haplotype; sample observations determine support.

Reuse sequence without changing the placement

```
CHROM=chr1 POS=200 ID=C SVTYPE=INS SVLEN=6
SEQ=>A@3A2H4=2H>2ITT
```

After unescaping, SEQ is **>A:2H4=2H>2ITT**: copy A[2:6] (GTTG), then add TT, giving **GTTGTT**. The shared piece may reuse A's graph segments, while C remains attached at its own chr1 breakpoint. Sequence reuse and parent placement are separate relationships.

Encoding	Meaning
>target:body / <target:body	Forward / reverse-oriented named target. Reverse matches use reverse-complement bases.
nH / n= / nD	Target clipping / exact copy / skip target bases. H contributes no allele bases.
nXbases / nIbases / nSbases	Literal mismatch / insertion / soft-clipped query bases.
>nIbases	Unnamed literal piece, e.g. >2ITT adds TT.

Merged FORMAT can use the implicit row representative: >8= is an eight-base match. Current rGFA sequence verification rejects ambiguous M operations; use explicit =/X or literal sequence. Do not apply INFO/SEQ decoding rules blindly to FORMAT. [3, 4]

05 / Sequence and provenance

Identify the sequence mode

Producing stage	Where the allele sequence lives
Default final caller	<code>##minsetrefSequenceEncoding</code> declares <code>SEQ=plain</code> and <code>EXTENDGRAPHCIGARPayload=absent</code> . Bare alignment operations do not preserve every alternate base.
Caller with <code>--seqcompress</code>	<code>SEQ=omitted</code> , inline CIGAR payloads; <code>INFO/SEQ</code> is a dot. This is not directly sufficient for the current rGFA insertion verifier, which requires <code>SEQ</code> .
Merged insertion output	<code>SEQ</code> may be literal or a graph encoding that references other allele IDs or sequence catalogs. Resolve all named dependencies.

Preserve the original assemblies when per-observation bases are needed. In rGFA mode, literal insertion blocks and SNP/SUB alleles are checked against eligible carrier assembly intervals. An alignment alone is not proof of sequence identity. [2, 4]

Decode values after splitting the table

Core escapes are `@40` for `@`, `@3A` for colon, `@3B` for semicolon, `@2C` for comma, `@09` for tab, and `@0A` for newline. Some packed compatibility payloads also use `@7C` for `|`. Decode `@40` last. These are project conventions; retain them when passing grVCF between LinGraph tools.

Keep catalogs with their identity and placement metadata

Preserve `##contig` lengths, `##alternativeLocus` source metadata, and the original per-sample `##pseudoLinearMapping` entries. Required data can include the calling reference, indexed query assemblies, the alternative/novel catalog, and freshly lifted local-reference templates. Changing the backbone requires compatible calls; renaming CHROM does not lift variants.

`--local-path-fasta` translates local VCF targets onto included roots using source interval, orientation, and sequence identity. `-a` includes every nonempty supplied record as a complete additional path. Do not substitute `-a` for `--local-path-fasta` when using a catalog of duplicated local loci.

Full-locus duplication

`PACCLASS=fulllocusdup` and `ALTERNATIVECIGAR` describe copies aligned to source loci. The exporter can share duplication paths for overlapping source intervals while keeping mismatches and insertions as literal branches. The merger drops these insertion calls unless `--keep-full-locus-dup-insertions` is used. [3, 4]

Recover SNPs inside inserted sequence

```
python tools/extract_insertion_snps.py \  
-i cohort.sv.vcf cohort.indel.vcf \  
--assemblies query_paths.txt -o insertion.snp.vcf -t 8
```

The extractor uses saved observation alignments and recursively follows named mappings to destination coordinates, including reverse mappings. Supply indexed assemblies for bare X mismatches. Inputs must contain the needed insertion definitions; do not append complete VCF files by concatenating headers. [5]

06 / Build and use a graph

Commands here use the GitHub repository layout: pipeline programs under scripts/ and utilities under tools/. Run them from the repository root with your actual data paths.

```
python scripts/merged_vcf_to_gfa.py \
-v cohort.sv.vcf cohort.indel.vcf cohort.snp.vcf \
-q query_paths.txt --graph-folder graph \
--local-reference-templates local_reference_templates.fa \
--local-path-fasta graph/summary/alternatives.fasta \
-o cohort.gfa -t 8
```

Pass completed files for the same backbone; -v also accepts a single combined VCF or repeated inputs. query_paths.txt contains **NAME FASTA [FAI]**, with names matching sample columns and relative paths resolved against the list directory. Local-reference templates must have current placement metadata for the selected backbone.

Option / output	Current behavior
Default	--gfa-mode rgfa and --all: SNPs, substitutions, insertions, deletions, and all sizes. General --size-cutoff defaults to 0.
--svonly --svcutoff 20	Only non-SNP structural events meeting the size cutoff, including qualifying deletions.
--insertion-only [SIZE]	Restrict to insertions; a bare flag uses 50 bp.
Backbone	Default: graph/inputs/reference_alternatives_novels.fa. Use -r plus --reference-haplotype when selecting a plain assembly backbone.
--validate-only	Extract/check sequence and resolve topology without writing GFA. --bed-only is coordinate output, not equivalent validation.
--drop-unverified	Opt-in omission of unverifiable variants and their dependents, with unresolved reports. Default is to stop.
--partition-records N	Bound conversion work by groups of whole dependent variant trees for large cohorts.
Graph and sidecars	rGFA S/L/P records, stable LN/SN/SO/SR tags, *.variants.tsv and anchor/mapping/verification reports.

Insertion/SNP/SUB P paths contain allele cores; deletion paths use flanks to bypass the deleted interval. Exact mapped runs reuse segments, and the parent/reference route remains present. This updates the older insertion-only, full-parent-path description. [4]

Assembly walks are a separate output

```
python scripts/gfa_sample_gaf.py -g cohort.gfa \
-v cohort.sv.vcf cohort.indel.vcf cohort.snp.vcf \
-s samples/*/*.vcf -q query_paths.txt -o gaf -t 8
```

Use the same merged inputs in the same order as graph export. GAF generation splices carried variants using original sample mappings; walks break at unresolved intervals or topology gaps. It is not a guarantee of complete chromosome-length phased paths. [6]

07 / Export standard VCF

For tools expecting reference-based VCF alleles, explicitly convert grVCF. Conversion chooses each row's representative; it does not retain all graph topology or expand every member allele. [7]

```
python tools/grvcf_to_vcf.py \
  -i cohort.sv.vcf -r reference.fa \
  -o cohort.sv.standard.vcf
```

Use the calling reference and adjacent .fai. Convert one input per invocation. Supply decoding catalogs with repeated **-a FASTA**; auxiliary catalogs do not add output reference contigs. Add **--normalize** for bcftools left alignment. BGZF .vcf.gz output and its CSI index also require bcftools.

Conversion result	What to expect
Explicit REF/ALT	SNPs, insertions, deletions, and replacements become base alleles with checked reference bases and indel padding.
Preserved record identity	IDs, sample names, genotypes, missingness, ploidy/phasing, QUAL, and FILTER are retained.
Standard annotations	Graph annotations are replaced with GT, AC, AN, AF, and NS. Genotypes support the chosen representative; member offsets and sequences are not expanded.
Graph-only records	Rows on contigs absent from the reference go to OUTPUT.unplaced.gr.vcf. Variants on insertion IDs generally belong here; they are not silently projected onto the chromosome.
Sequence resolution	Named allele/FASTA dependencies and forward references are resolved. Missing bases, ambiguous M, conflicting IDs, or reference mismatches cause an error.

What to share with another user

Send the grVCF with its complete headers, the reference/backbone identity, required indexed sequence catalogs, and the producer version/options. Include original sample VCFs and assembly paths when downstream work needs sample walks or missing query bases. Keep standard VCF and its unplaced grVCF together when sharing a reference-based export.

Implementation sources

Reviewed 4 October 2026. Core caller, merger, nested merger, and rGFA modules match GitHub main commit **227d42e6ac5d**. Utility copies were also checked against the repository layout. This guide is based on implementation behavior where older prose documentation differs.

[1] [VCF envelope reference](#)

[2] [Caller schema and mapping metadata](#)

[3] [Merged fields and nested events](#)

[4] [rGFA parser, allele metadata and coordinate resolver](#)

[5] [Insertion SNP extraction](#)

[6] [Sample GAF generation](#)

[7] [Standard VCF conversion](#)

[4] Also consult scripts/gfa_interval_metadata.py, gfa_stable_coords.py, gfa_interval_pipeline.py, and gfa_topology.py at the same source revision.